GAN Image Generation Research

How to Solve Mode Collapse in Deep Learning

Overview

This project uses **GAN (Generative Adversarial Network)** to generate images. The concept is simple: two AI models compete against each other — one creates "fakes" and the other tries to detect them. Eventually, the faker becomes increasingly skilled.

3	100	29%	64px
Versions	Epochs	Improvement	Resolution

The Problem

What is Mode Collapse?

The most common issue when training GANs is "Mode Collapse":

> The Generator discovers that producing only one specific type of image can fool the Discriminator. So it takes a "shortcut" and only produces that one type. The result: **all generated images look exactly the same**.

What Happened in My Experiment?

In my first experiment, Mode Collapse suddenly occurred at epoch 91:

- All generated truck images became the same color
- Almost no variation between images
- Discriminator accuracy spiked to nearly 100%

Root Cause: The Discriminator became too strong, leaving the Generator unable to learn, forcing it to find shortcuts.

Solutions

I experimented with 3 versions, progressively solving the problem:

V0: Baseline Failed

Training all 10 CIFAR-10 classes → Mode Collapse at epoch 91

V1: Single-Class Training Success

Only training on "truck" class, with Label Smoothing to "weaken" the Discriminator

V2: Adding Dropout Best

Added Dropout to Discriminator, further preventing it from being too strong → 29% improvement

Key Techniques

1. Label Smoothing

2. Dropout Regularization

Changed "real=1, fake=0" to "real=0.9, fake=0.1" to prevent Discriminator overconfidence

Randomly disable 30% of neurons to prevent Discriminator from overfitting

3. Training Balance Control

When Discriminator accuracy >80%, reduce its training frequency

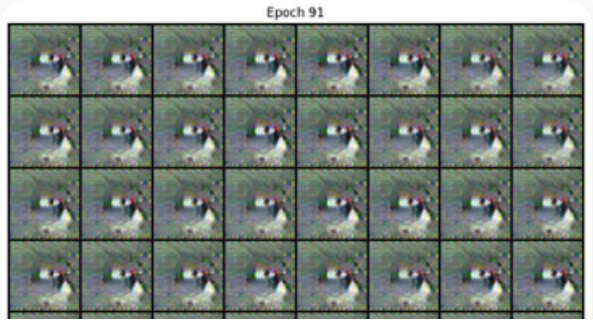
4. Single-Class Training

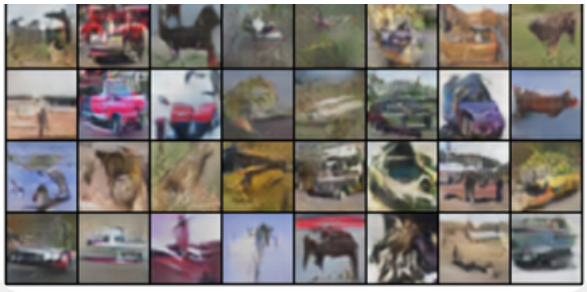
Start with a single class to reduce learning complexity

Results

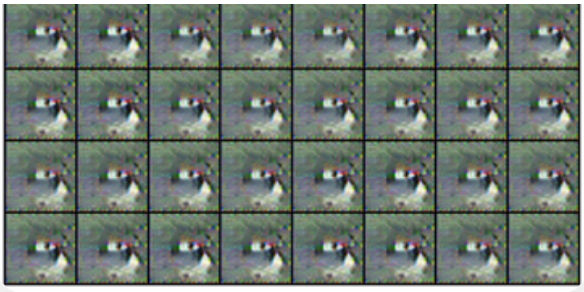
Mode Collapse Comparison (V0 vs V1/V2)

V0 Mode Collapse (Epoch 91)





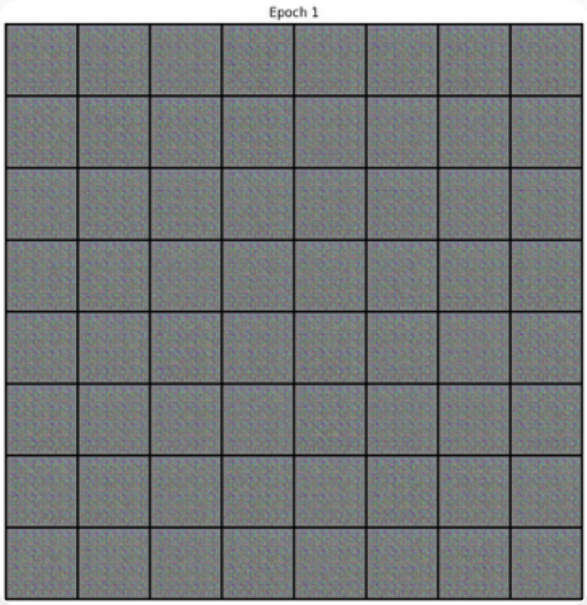
V0 Epoch 65 - Still Normal



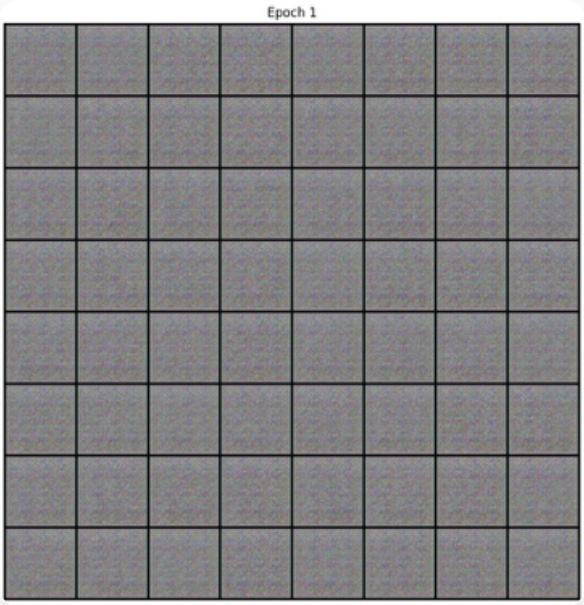
V0 Epoch 91 - Mode Collapse!

Training Progress Comparison (V1 vs V2)

Early Training (Epoch 1)

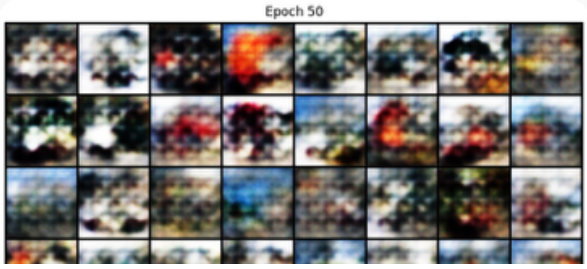


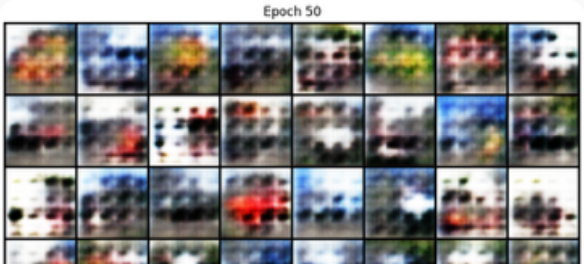
V1 - Baseline

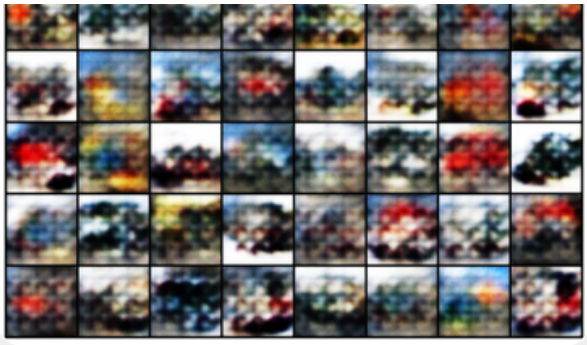


V2 - With Dropout

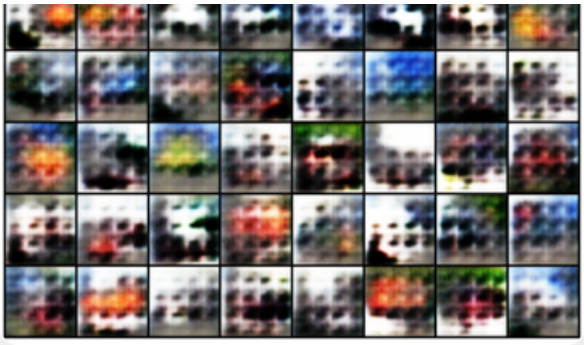
Mid Training (Epoch 50)





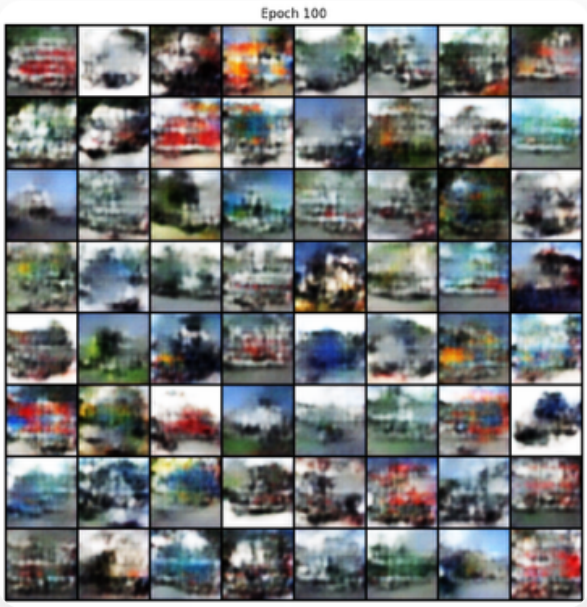


V1 - Baseline

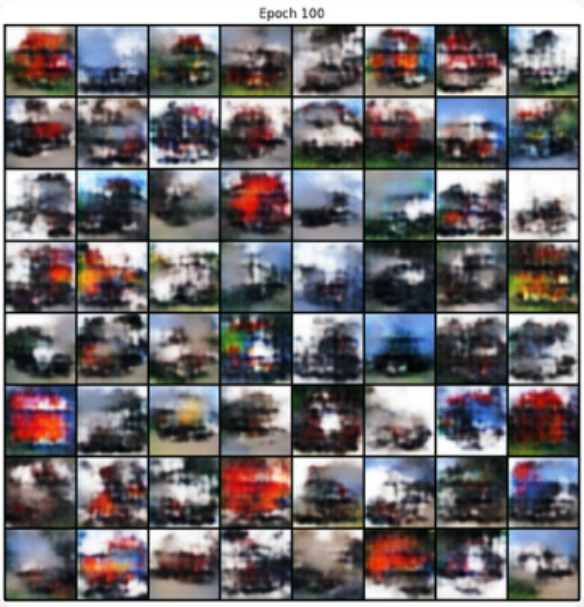


V2 - With Dropout

Training Complete (Epoch 100) - Final Results

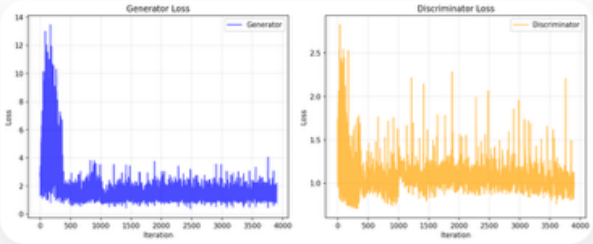


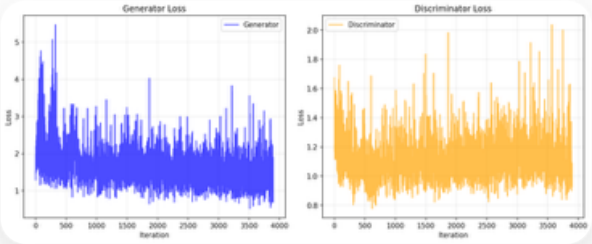
V1 - Baseline



V2 - With Dropout (Clearer, More Diverse)

Training Loss Curves





V1 Training Curve

V2 Training Curve

How to Read These Charts?

Each chart has two lines: **Blue is the Generator**, **Orange is the Discriminator**.

Healthy Training

The two lines should "pull and push" each other, maintaining dynamic equilibrium. If one keeps dropping while the other keeps rising, it indicates imbalance.

Why is V2 Better?

V2's curves have smaller fluctuations and are more stable, indicating smoother training with better balance between Generator and Discriminator.

Conclusion

Balance is Key

The key to GAN training is keeping the Generator and

Start Simple

Begin with a single class, then tackle more complex tasks

Discriminator balanced. If either becomes too strong, problems arise.

after achieving success.

About Image Quality

While the results "look like something," the images are still **very unclear** and hard to recognize. This is not just a training issue — the CIFAR-10 dataset itself has very low resolution (originally only 32×32), so even upscaled to 64×64, the details are inherently limited.

Future Plans: Experiment with Diffusion Models and higher resolution image datasets to compare the effectiveness of different generative models.